Spiro

A low-cost 3D-printed IoT spirometer with a disposable mouthpiece designed from burned-field crop residue

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Team MEDTHAI
Puttitada Pruetchayakul (leader) · Pimnara Roongsetthavitr
Suchanan Kittikraisak · Natnicha Taepucharoen · Nabhat Sethaudom
Advisor: Tanachat Srinurat · Thailand

1 · Problem

Smoke from burning rice straw and cane leaves carries PM2.5 deep into the lung, linked to cough, asthma and COPD.

- COPD: ~3.4 million deaths in 2023, ~6% of all deaths (WHO)
- Pathumthani survey (n=1,014): COPD 8.3%, but only 9.5% of them had been diagnosed
- Open burning of crop residue: ~524 billion baht/year damage in Thailand

Spirometry is the confirming test, but it stays where the machine and budget are.

2 · Our idea

A handheld spirometer a school or village health post can build, with results sent to a doctor by phone.

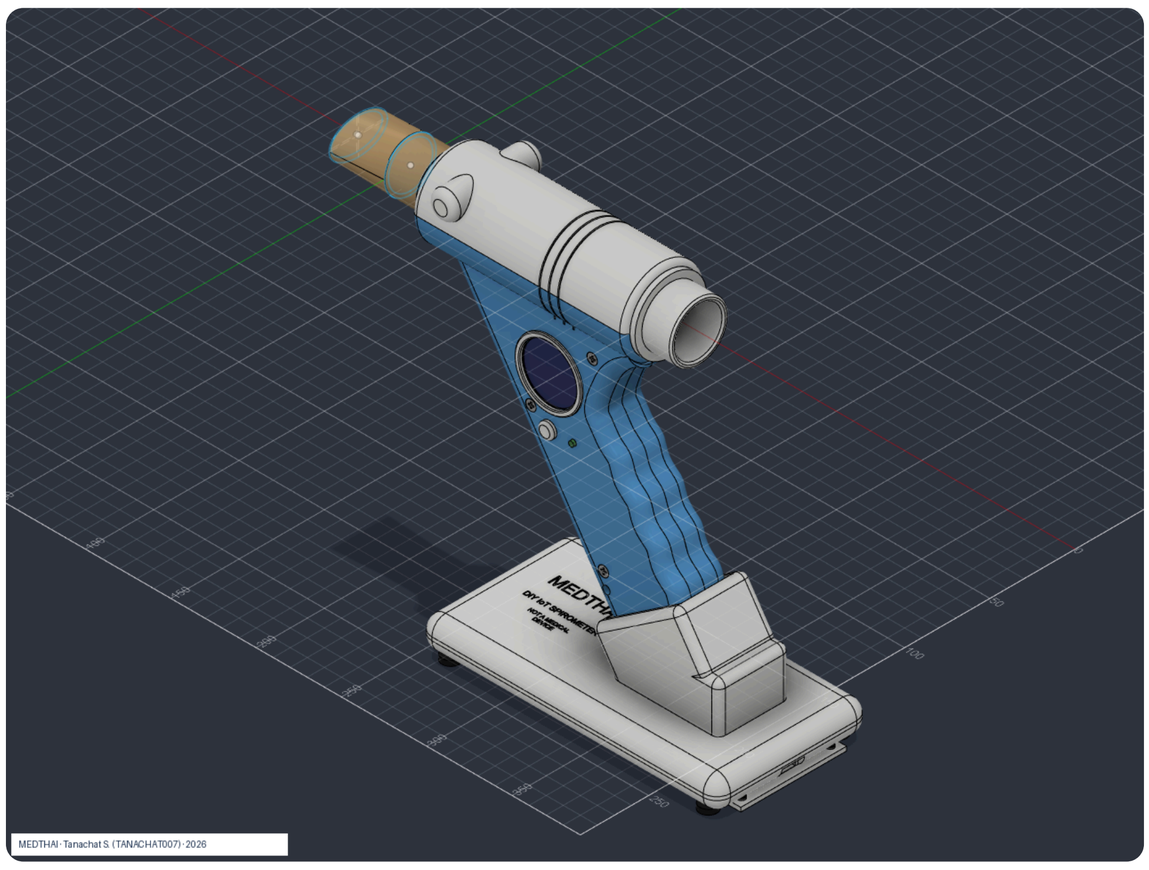
- **Eco-friendly:** single-use mouthpiece planned from rice straw / cane leaves
- **Environment + health:** the fields that cause the problem supply the material to screen it
- **Affordable:** ~3,100 baht (~USD 85) in parts, 26 items

3,084฿ estimated BOM

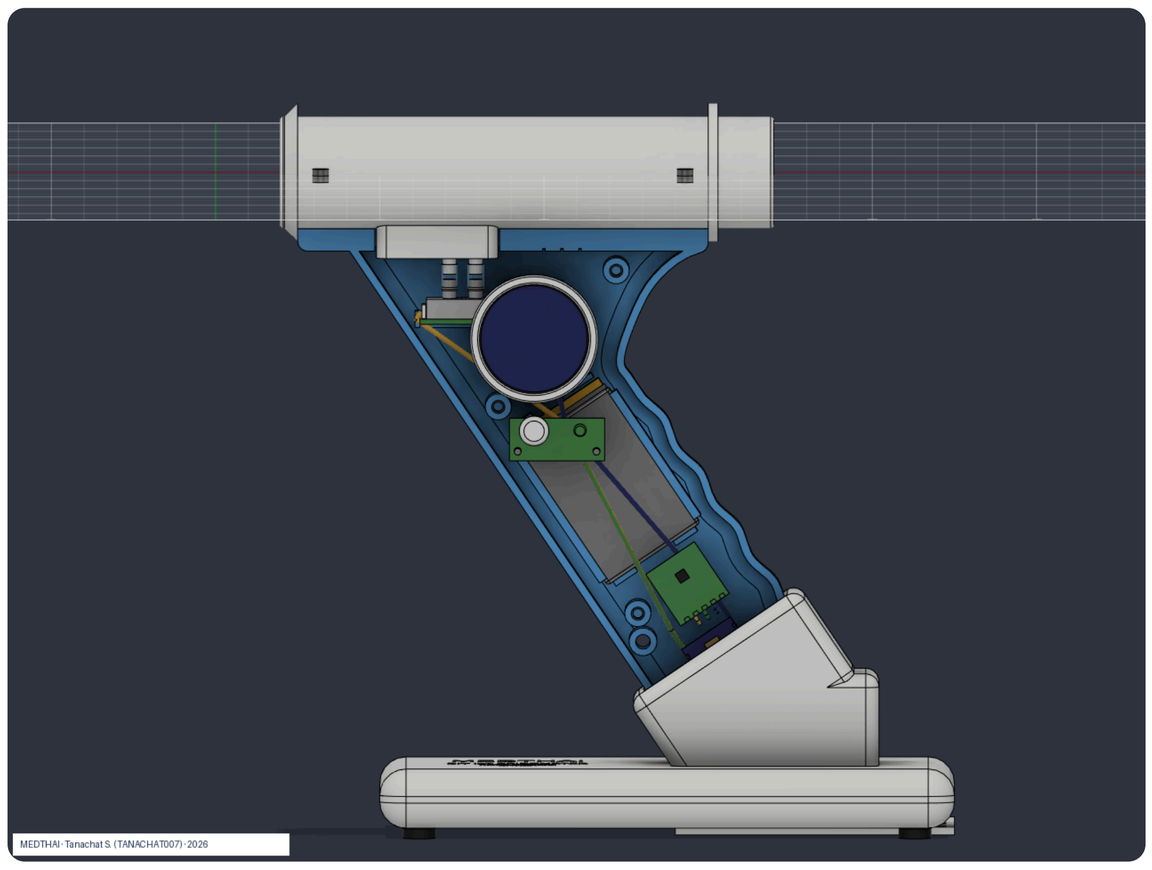
10 printed PETG parts

400 Hz sensor sampling

3 · Design (version 3)



CAD render: handheld unit on magnetic charging dock



Internals: sensor, round display, battery, ESP32-S3

- Printed venturi (28 mm → 23 mm throat) + Sensirion SDP810-500Pa; Q = K·√ΔP, integrate for volume
- Reports FVC, FEV1, FEV1/FVC and PEF; XIAO ESP32-S3 with 1.28" round display, speaker prompt, BLE + Wi-Fi
- Washable venturi tube; replaceable mouthpiece counted by firmware (reminder at 20 blows / 7 days)

4 · Mouthpiece from burned fields

Stage	Material
Prototype	PETG print (fit check)
Production	Moulded rice-straw / cane-leaf pulp (proposal)
Fallback	Standard cardboard mouthpiece

Straw trays (up to 80% straw) beat polystyrene in tension and lost ~20% mass in 4 weeks in soil. A mouthpiece is not a filter: a standard disposable filter is part of the plan.

5 · Online system

- **Patient:** register, blow or follow coaching, history and trend
- **Doctor** (admin-approved): clinic mode, ≥3 blows, repeatability 0.15 L, live curve, A4 report
- **Admin:** approvals, audit log
- Firebase Hosting + Auth + Firestore; rules tested incl. privilege-escalation attempts
- Voice-guided nurse-style training script (Thai / English)

6 · Evidence

- Printed venturi + ESP32: ~1.5% error vs 2 L pump
- Venturi home spirometer: r = 0.992 (FVC), 0.968 (FEV1)
- Frugal printed venturi, 30 people: ICC 0.97 for FEV1 (peak flow less accurate)

Confidence: HIGH measuring principle · MEDIUM straw mouthpiece · LOW-MEDIUM infection control without filter.

7 · Honest limits

- Version 3 not yet printed, assembled or measured; K needs 3 L syringe calibration
- Firmware tail correction (FVC may read ~12% low without it) is unvalidated
- No clinical study, no regulatory approval. **Education and screening only, not for diagnosis.**

8 · Next steps and impact

Calibrate with a 3 L syringe → compare 20+ volunteers with a hospital spirometer under supervision → mould and test pulp mouthpieces (resistance, leak, wet strength) → publish results whether favourable or not. Built on our SmartLung platform (WICO 2026 Gold Award, iCAN 2026 Silver Medal; awards were for SmartLung, not for a spirometer).

Sources: WHO COPD fact sheet; Pathumthani spirometry survey; Frontiers Bioeng. Biotechnol. 2025 (printed venturi); Bangor Univ. straw-tray study; full reference list in the essay. Contact: Team MEDTHAI · Demo: medthai-spirometer.web.app